

Proposition (1)

L/k : extension
 α, β : algebraic / k

Then the following elements are all algebraic over k :

(i) $\alpha + \beta$

(ii) $\alpha\beta$

(iii) $(\alpha \neq 0) \alpha^{-1}$

Pf: α is algebraic $\Leftrightarrow k(\alpha)/k$ is finite

& $\alpha^{-1} \in k(\alpha)$

$\Rightarrow \alpha^{-1}$ is algebraic over k .

$k \leq k(\alpha, \beta) \leq L$
 $\downarrow$
 $\alpha, \beta, \alpha\beta, \alpha+\beta$

WTS: $k(\alpha, \beta)/k$ is finite

Observation ①

$$k(\alpha, \beta) = (k(\alpha))(\beta) \leq L$$

$$\underbrace{k}_{\text{finite}} \leq k(\alpha) \leq \underbrace{k(\alpha)(\beta)}_{\text{finite}}$$

$\Rightarrow k(\alpha, \beta) = k(\alpha)(\beta)$ is a finite extension of k

□

Embeddings of field extensions

$$L/k, \quad (k[x]/(f(x)))/k, \quad f(x) \in k[x] \text{ irreducible}$$

$$\text{Emb}_k \left(k[x]/(f(x)), L \right)$$

$$k[x]$$

$$\begin{array}{ccc} & \searrow \text{ev}_\alpha : f(\alpha) = 0 & \\ & & \downarrow \\ k[x]/(f(x)) & \xrightarrow{\text{ev}_\alpha} & L \end{array}$$

$$\text{Emb}_k(k[x]/(f(x)), L) \xrightarrow{\sim} \{\alpha \in L : f(\alpha) = 0\}$$

Observation (2)

$$\left| \text{Emb}_k(\underbrace{k[x]/(f(x))}_M, L) \right| \leq \deg f(x)$$

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$[M:k]$

Example: $k = \mathbb{Q}$

$$f(x) = x^3 - 2 \in \mathbb{Q}[x]$$

$$M = \frac{\mathbb{Q}[x]}{(x^3 - 2)}$$

$$\text{Emb}_{\mathbb{Q}}(M, \mathbb{R}) \cong \{\sqrt[3]{2}\}$$

$$\text{Emb}_{\mathbb{Q}}(M, \mathbb{C}) \cong \left\{ \sqrt[3]{2}, e^{2\pi i/3} \sqrt[3]{2}, e^{4\pi i/3} \sqrt[3]{2} \right\}$$

Proposition (2) : L/k , M/k
finite

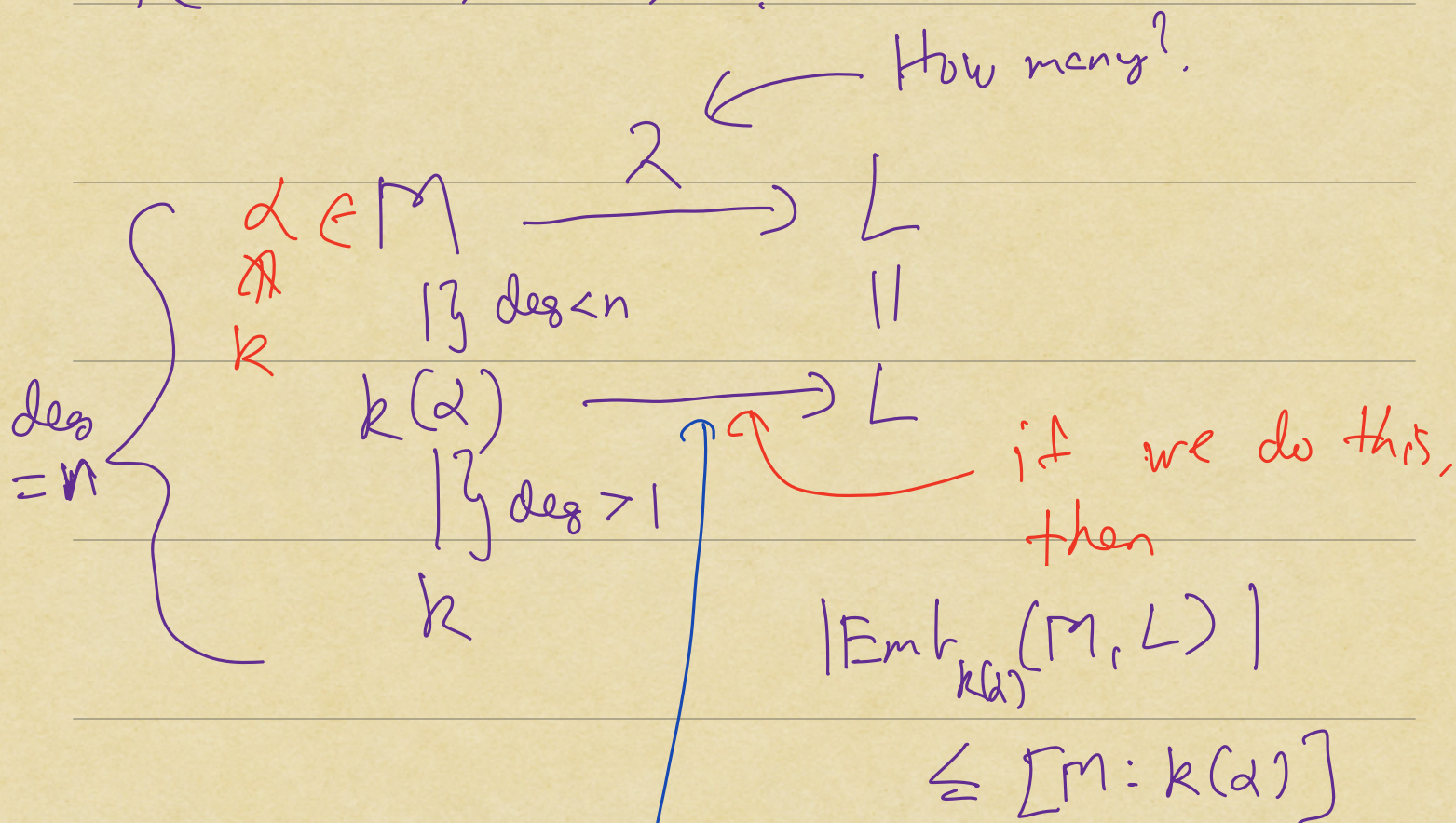
$$|\text{Emb}_k(M, L)| \leq [M:k]$$

Pf: Proof will be by induction
on degree $[M:k]$

Statement: Proposition is true for
 $P(n)$ extensions $L/k, M/k$
with $[M:k] \leq n$.

$P(1) \checkmark$

$P(n-1) \Rightarrow P(n) ?$



of possibilities
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$$|\text{Emb}_k(\alpha, L)| \leq [k(\alpha):k]$$

$$\Rightarrow |\text{Emb}_k(M, L)| \leq [k(\alpha):k] [M:k(\alpha)] \\ = [M:k]$$